

Comparing venture capital syndication and its outcomes in the USA and China: an institutional theory perspective

Journal of
Entrepreneurship
in Emerging
Economies

1511

KwangWook Gang

*Department of Management, Franklin P Perdue School of Business,
Salisbury State University, Salisbury, Maryland, USA*

Yonggi Kim

*Division of Global Innovation Strategy Research,
Science and Technology Policy Institute, Sejong, Republic of Korea*

Byunghoon Kim

*Department of Industrial and Management Engineering,
Hanyang University ERICA, Ansan, Republic of Korea, and*

Seungbeom Kim

College of Business Administration, Hongik University, Seoul, South Korea

Received 17 April 2025
Revised 9 June 2025
Accepted 10 June 2025

Abstract

Purpose – This study aims to explore how institutional environments shape the effectiveness of venture capital (VC) syndication by comparing the USA and China – representative cases of rule- and relationship-based systems, respectively. VC syndication is often touted for enabling resource pooling, risk sharing and performance-enhancing collaboration, yet its outcomes may vary significantly depending on institutional constraints. This study investigates whether syndication – especially global syndication and investor experience – improves startup exit performance across contrasting governance contexts. By doing so, this study contributes to institutional theory by examining the boundary conditions under which syndication strategies enhance or hinder entrepreneurial outcomes in deep-tech industries like semiconductors.

Design/methodology/approach – Using a unique data set from CB Insights, the authors analyze 11,545 VC investment rounds across 1,391 startups in the semiconductor sectors of the USA and China. They develop institutionalized hypotheses linking syndication features to startup exit probabilities and use logistic regression and IV-Probit models to assess robustness and address endogeneity concerns. The models are estimated separately by country to capture institutional variation. They examine both structural aspects of syndication – such as syndicate size and foreign investor participation – and relational characteristics, such as average and variance of investor experience, to evaluate how these factors interact with national institutional contexts.

Findings – The results highlight the contingent value of VC syndication. In the USA, global syndication is positively associated with startup exit success, reflecting the role of formal institutions in facilitating cross-border collaboration. However, syndicate size and investor experience show limited or inconsistent effects. In contrast, in China, both domestic and global syndication are negatively related to exit outcomes, likely due to coordination difficulties and institutional frictions. Investor experience has mixed effects, with closed guanxi networks sometimes constraining broader strategic support. These findings suggest that syndication's benefits are not universal but depend heavily on institutional fit and trust mechanisms.



Funding: This study was supported by Ministry of Education of the Republic of Korea and the National Research Foundation of Korea, Grant No. NRF-2024S1A5C3A02043476.

Journal of Entrepreneurship in
Emerging Economies
Vol. 17 No. 6, 2025
pp. 1511-1532
© Emerald Publishing Limited
2053-4604
DOI 10.1108/JEEE-04-2025-0198

Originality/value – This study advances institutional theory by identifying how VC syndication outcomes depend on the alignment between syndicate structure and institutional context. The authors introduce the concept of institutional friction as a moderating force that can invert the expected benefits of syndication in relationship-based economies. The findings highlight that global syndication enhances success in rule-based settings but may underperform where informal governance dominates. The research offers actionable insights for policymakers and practitioners by showing that syndication strategies must be tailored to institutional realities. It also contributes empirically to the understudied intersection of venture finance, institutional voids and cross-border entrepreneurial ecosystems.

Keywords Venture capital syndication, Institutional theory, Startup exit, Global investment, US–China comparison, Semiconductor industry

Paper type Research paper

1. Introduction

Venture capital (VC) syndication – where multiple investors jointly fund startups – is a well-established strategy to share risk, pool resources and enhance access to expertise (Brander *et al.*, 2002; Dimov and Milanov, 2010; Kim and Park, 2021). While syndication is globally practiced, its effectiveness depends on the institutional environment in which it operates. The USA and China, the world's two largest economies, represent contrasting institutional settings that influence how VC syndication is structured, coordinated and translated into entrepreneurial success.

Institutional theory provides a useful framework for comparing these contexts. In the USA, formal institutions such as contract law, investor protections and enforcement mechanisms support transparent, scalable syndication (Scott, 2013; Hochberg *et al.*, 2007). In contrast, China's institutional framework is characterized by state involvement, limited legal enforcement and reliance on informal relational networks (Peng *et al.*, 2008; Bruton and Ahlstrom, 2003). These differences raise important questions about whether syndication – and especially global syndication – enhances or impedes startup outcomes in varying institutional contexts. While cross-border VC flows are growing, little research has systematically compared syndication structures across institutional regimes. Prior studies primarily focus on domestic syndication (Gompers and Lerner, 2004; Brander *et al.*, 2002), leaving open questions about how institutional environments shape the logic and consequences of both local and global syndicates.

This study examines how institutional conditions influence VC syndication and investor dynamics in the USA and China's semiconductor sectors. We draw on dynamic institutional theory (Aeeni *et al.*, 2019, 2024) to view institutions not merely as constraints, but as evolving structures coshaped by entrepreneurial activity. This lens allows us to explore how VCs adapt to or overcome institutional limitations, and whether these adaptations generate differential exit outcomes. We aim to: compare syndication practices between the USA and China, evaluate how VCs manage institutional constraints and assess how syndication features (e.g. investor experience, global participation) affect exit success. In doing so, we contribute to institutional theory by specifying how syndication outcomes are conditioned by institutional fit – and potentially reversed in relationship-based contexts.

2. Theoretical background

2.1 Institutional theory and entrepreneurship

Institutional theory provides a robust framework for understanding how formal and informal structures shape entrepreneurial behavior. Formal institutions – such as laws, regulations and enforcement mechanisms – establish the rules of market engagement and ensure transparency, while informal institutions – including cultural norms, trust and social

networks – play a critical complementary role, particularly in environments where formal systems are weak or inconsistently applied (North, 1990; Scott, 2013; Peng, 2003).

Organizations gain legitimacy by aligning their practices with the expectations and constraints imposed by these institutional environments (Meyer and Rowan, 1977). Legitimacy, in turn, facilitates access to critical resources such as funding, talent and partnerships. However, institutional environments are not static. Recent work has highlighted how entrepreneurial actors actively navigate, reinterpret and even reshape institutional conditions – a perspective sometimes referred to as the “entrepreneur–institution nexus” (Aeeni *et al.*, 2019, 2024).

Bruton *et al.* (2010) categorize the institutional research on entrepreneurship into three core streams: the institutional setting and its impact on entrepreneurship, the importance of legitimacy and the concept of institutional entrepreneurship. First, institutional environments can enable or constrain entrepreneurial activity. In supportive contexts – with accessible capital, favorable regulation and strong market infrastructure – entrepreneurship tends to flourish (Foster, 1986; Gnyawali and Fogel, 1994). Conversely, in weak institutional contexts marked by regulatory opacity or corruption, entrepreneurship may be stifled or diverted into unproductive channels (Baumol *et al.*, 2007; Soto, 2013). Governments play a central role in shaping institutional contexts. Effective states reduce barriers to entry, address market failures and encourage efficient resource allocation through policy design (Baumol, 1996). In many emerging economies, however, the state may fail to enforce property rights or legal contracts, forcing entrepreneurs to rely on informal networks and social capital to navigate market activities (Bruton and Ahlstrom, 2003). While informal institutions can substitute for formal ones in some cases, they may also increase transaction costs and introduce inefficiencies (Rajan and Zingales, 2003).

Second, legitimacy is vital to new ventures. Because early-stage firms often lack a track record of performance, gaining legitimacy becomes essential for securing external support (Stillman, 1974). Institutional theory identifies three pillars of legitimacy: regulatory (laws and rules), normative (social values and norms) and cognitive (shared beliefs and mental models) (Scott, 2013). These dimensions interact to determine how stakeholders perceive and support new ventures. Entrepreneurs in emerging markets must navigate institutional voids by crafting legitimacy strategies suited to their specific institutional context, such as aligning with dominant cultural norms or partnering with reputable institutions (Ahlstrom *et al.*, 2008; Bruton *et al.*, 2009).

Third, institutional entrepreneurs proactively shape their institutional environment to create conditions favorable to venture success (DiMaggio, 1988). These actors may establish new norms, push for regulatory changes or create governance structures that facilitate innovation and coordination (Garud *et al.*, 2002). Especially in under-organized or transitioning markets, entrepreneurs often serve as catalysts for institutional transformation (Lawrence *et al.*, 2002; Rao *et al.*, 2000).

2.2 Institutional theory and venture capital decisions in the global context

Institutions play a critical role in shaping the decision-making of venture capitalists (VCs). While VC investment logic tends to converge around certain principles globally due to institutional isomorphism (DiMaggio and Powell, 1983), how information is interpreted and acted upon varies significantly across institutional settings (Sapienza and Korsgaard, 1996). Institutional theory helps explain why syndication structures, investor preferences and expected outcomes differ across countries.

Institutional environments are often categorized into rule- and relationship-based systems (Peng, 2003). Rule-based economies like the USA rely on strong formal institutions that enforce contracts and protect property rights. These reduce uncertainty and facilitate

transactions between parties who may not share prior relationships. Within such contexts, VCs often prioritize scalable investments, rigorous due diligence and market-based signals (Zacharakis and Meyer, 1998). The US venture capital industry exemplifies this model, with mature legal systems and robust enforcement mechanisms that support efficient syndication (Hall and Hofer, 1993).

Conversely, relationship-based economies such as China operate in environments where formal institutions are weaker or under development. In such settings, trust is built through personal connections, repeated interactions and reputation rather than contracts. VCs in these contexts often rely on relational networks to mitigate risks and access opportunities (Williamson, 1985; Bruton and Ahlstrom, 2003). While this system fosters close-knit collaboration, it can also limit scalability and increase coordination challenges.

China presents an especially compelling case. Its VC industry has grown rapidly within a transitional institutional landscape. Despite improvements, regulatory barriers and weak enforcement remain significant. Investments involving state-owned enterprises (SOEs) further complicate governance and strategic alignment (Folta, 1999; White *et al.*, 2002). These dynamics affect not only where and how VCs invest but also how they engage in syndication – particularly with global partners.

In contrast, the US VC ecosystem, rooted in post-Second World War economic development, has built extensive institutional infrastructure supporting innovation and investment (Bygrave *et al.*, 2022). Formal contracts and enforcement mechanisms enable syndication with minimal friction, allowing investors to focus on strategic fit and resource complementarities rather than interpersonal trust.

2.3 Venture capital syndication

VC syndication is a widespread practice in which multiple investors pool capital, share expertise and manage risk jointly. It facilitates better decision-making by incorporating diverse perspectives, helps reduce financial exposure and enhances access to future opportunities through network expansion (Brander *et al.*, 2002; Sorenson and Stuart, 2001). Syndication also acts as a governance mechanism, promoting accountability and reducing opportunistic behavior among participants (Poppo, Zhou, and Rhu, 2008). Syndicates benefit from resource pooling and access to complementary skills – particularly valuable in high-tech or cross-border ventures where information asymmetry is high (Tian, 2011; Cumming and MacIntosh, 2006). Furthermore, syndication provides informational and reputational advantages, improving monitoring and post-investment support (Manigart *et al.*, 2006). Yet, syndication is not without its challenges. Coordination becomes increasingly complex with the size and heterogeneity of the syndicate. Differences in risk tolerance, governance styles or exit strategies can create conflict and delay decision-making (George *et al.*, 2016; Heidl *et al.*, 2014). These difficulties are exacerbated in environments with weak formal institutions, where informal coordination must compensate for a lack of enforcement.

In large syndicates, such tensions are magnified. Diverse investor backgrounds may bring valuable knowledge but also lead to strategic misalignment. Research suggests that the relationship between syndicate size and performance is nonlinear, with both benefits and drawbacks depending on context (Falconieri *et al.*, 2019; Kim and Park, 2021). By examining syndication practices in both the USA and China, this study explores how institutional differences shape the formation, coordination and outcomes of venture capital syndicates. We consider how investors respond to institutional constraints and how experience, partner diversity and international participation affect exit success in rule- versus relationship-based economies.

3. Hypothesis development

3.1 *Venture capital syndication counts and venture performance between the USA and China*

VC syndication can benefit both VC firms and companies in need of financial resources. VC firms vary in the amount of available capital for investment opportunities, as well as in their skills, experiences, networks and knowledge, which depend on the scope of their investment history. The advantages firms offer may include organizing production, establishing connections with potential business partners and customers and recruiting key human capital (Hochberg *et al.*, 2015). A large syndicate could also improve the chances of a venture securing the necessary capital in subsequent funding rounds (Zhang *et al.*, 2017).

In rule-based economies like the USA, institutional frameworks emphasize strong legal and regulatory systems that ensure contract enforcement and property rights protection (North, 1990; Peng, 2003). These frameworks reduce transaction costs and foster collaboration, making syndication an effective strategy for VCs to pool resources and expertise. In the USA, trust is built through legal frameworks rather than personal relationships, enabling syndicate members to work cohesively without relying on prior connections (Williamson, 1985). Syndication allows VCs to combine financial and managerial resources, enhance due diligence and improve monitoring, which increases the scalability of portfolio companies and their readiness for successful exits, such as initial public offerings (IPOs) or acquisitions (Gompers and Lerner, 2001). The US VC market also prioritizes market-based factors such as market growth and competitive advantage, which syndication can support by enhancing expertise and networks (Hall and Hofer, 1993; Kim and Park, 2021; Zacharakis and Meyer, 1998). Therefore, we hypothesize that increased syndication will increase the probability of successful exits in rule-based economies like the USA.

In contrast, relationship-based economies like China prioritize trust built through personal relationships rather than formal legal systems (Peng, 2003). While this fosters close-knit collaborations within local networks, it complicates syndication, which requires effective coordination among multiple parties. The weaker enforcement of property rights and contracts in China increases transaction risks in syndication (Bruton and Ahlstrom, 2003; Folta, 1999). Moreover, the need for strong relational trust complicates managing multiple relationships within a syndicate, as it dilutes the personalized trust that is key to collaboration in such economies (Lasserre and Probert, 1994). Chinese VCs often focus on investments in SOEs, which may create strategic misalignments within syndicates and reduce the effectiveness of syndication. Without robust legal frameworks, the increased complexity of syndication can lead to inefficiencies and reduced performance. Thus, we hypothesize that increased syndication will decrease the probability of successful exits in relationship-based economies like China:

H1a. VCs in strongly rule-based economies (e.g. the USA) will exhibit a higher probability of successful exits as the number of domestic VC syndication increases.

H1b. VCs in strongly relationship-based economies (e.g. China) will exhibit a lower probability of successful exits as the number of domestic VC syndication increases.

3.2 *Global venture capital syndication*

International VC syndication plays a critical role in transferring expertise from developed to emerging markets. Domestic VC firms, particularly in underdeveloped regions like Asia,

benefit significantly from partnerships with experienced foreign firms (Dai *et al.*, 2012; Wright and Lockett, 2003). Asian VC markets, characterized by high information asymmetry and limited early-stage investment, lag in expertise compared to North America and Europe (Bruton *et al.*, 2005). Foreign partners bring specialized knowledge in portfolio selection, high-tech investments and disinvestment strategies, enabling domestic firms to learn and adapt (Cumming and Dai, 2010). These learning opportunities allow domestic VC firms to develop skills for investing in high-risk, high-reward sectors such as technology. While most Asian VC investments traditionally focus on later-stage and non-technology ventures (Dai *et al.*, 2012), exposure to international syndicates fosters expertise in early-stage, high-tech ventures. This shift enhances their ability to address agency problems and diversify investment strategies effectively (Jensen and Meckling, 1976). Syndication also improves investment performance. Research highlights that joint ventures between foreign and domestic VC firms often lead to better exit outcomes (Lerner, 1994; Megginson and Weiss, 1991). The transfer of knowledge through syndication improves decision-making and exit strategies, resulting in higher success rates (Wang and Wang, 2011). Consequently, domestic VC firms with international syndication experience are more likely to invest in high-risk ventures and achieve successful exits (Khurshed *et al.*, 2020).

The US venture capital industry operates within a mature, rules-based institutional framework characterized by strong legal protections and efficient enforcement mechanisms (North, 1990). These features reduce transaction costs and uncertainty, enabling VCs to engage effectively in global syndication. Global syndication allows US VCs to diversify investments, access larger networks and leverage partners' expertise to maximize successful exits. In rule-based economies like the USA, syndication benefits from institutionalized trust and impersonal legal enforcement (Peng, 2003). Furthermore, the focus on market-based factors such as scalability and competitive advantage enhances the effectiveness of global syndication (Hall and Hofer, 1993; Zacharakis and Meyer, 1998). Thus, increased global syndication improves collaboration and exit probabilities for US VCs.

In contrast, China's relationship-based environment prioritizes personal trust and connections in investment decisions (Peng, 2003). This reliance on relational dynamics complicates global syndication, as trust must be built through long-term interactions, which are difficult to establish in international partnerships. The weak legal and regulatory frameworks in China further exacerbate this challenge, as weak enforcement mechanisms increase risk and limit the scalability of transactions with foreign partners (Bruton and Ahlstrom, 2003; Folta, 1999). Chinese VCs may struggle to leverage global syndication effectively, limiting their ability to achieve successful exits through international collaboration. Furthermore, Chinese VCs' focus on SOEs misaligns with the fast-paced, high-growth ventures typically targeted by global syndicate partners (White *et al.*, 2002). Consequently, we hypothesize that global syndication will decrease the probability of successful exits in relationship-based economies like China:

- H2a. VCs in strongly rule-based economies (e.g. the USA) will exhibit a higher probability of successful exits as global VC syndication increases.
- H2b. VCs in strongly relationship-based economies (e.g. China) will exhibit a lower probability of successful exits as global VC syndication increases.

3.3 The nature of venture capital syndication

Investor experience plays a critical role in VC syndication outcomes (Khurshed *et al.*, 2020). Experienced investors bring valuable expertise, networks and strategic insights that enhance

due diligence, governance and portfolio company scaling – factors that collectively increase the likelihood of successful exits such as IPOs or acquisitions (Gompers and Lerner, 2001; Jensen and Meckling, 1976). Their industry knowledge and reputational capital help ventures navigate challenges, attract acquirers and position themselves competitively (Harrison and Mason, 2000; Hellmann and Puri, 2002). Syndicates with greater aggregate experience benefit from pooled expertise, promoting better decision-making and reducing agency risks (Brander *et al.*, 2002).

The effect of investor experience, however, is shaped by the institutional context. In rule-based economies like the USA, where legal systems ensure transparency and contract enforcement, experienced investors can focus on strategic oversight rather than compensating for institutional voids. Their ability to guide governance and risk management is amplified by a formal environment that supports scalable collaboration (North, 1990; Hall and Hofer, 1993; Zacharakis and Meyer, 1998). In contrast, in relationship-based economies like China, trust is built through long-term relationships rather than legal contracts (Peng, 2003). Here, experienced investors serve a different function: leveraging social capital to navigate informal governance, build trust, and secure resources (Williamson, 1985). Their embeddedness in local networks helps overcome institutional weaknesses and supports exit outcomes, especially in opaque or state-influenced markets (Fukuyama, 1996). Thus, while the mechanisms differ, investor experience enhances exit success in both institutional contexts by providing essential resources and legitimacy:

H3. In both strongly rule-based economies (e.g. the USA) and relationship-based economies (e.g. China), VCs will exhibit a higher probability of successful exits as the experience of same-round investors increases.

The variance in the experience of same-round investors can have profound effects on the likelihood of successful exits, particularly in relation to the institutional environment. In rule-based economies like the USA, where legal protections and formalized regulatory systems are robust (North, 1990), a higher variance in investor experience can significantly enhance decision-making and governance. A syndicate with investors who bring varying levels of experience to the table is likely to benefit from diverse perspectives, strategic insights and complementary skill sets. More experienced investors contribute a deep understanding of market trends, regulatory environments and exit strategies, while less experienced investors may bring fresh ideas and new approaches to risk management, potentially enhancing innovation and operational efficiency (Harrison and Mason, 2000). This diversity in experience facilitates a more comprehensive evaluation of investment opportunities and risks, leading to better-informed decisions and stronger governance (Gompers and Lerner, 2001).

Furthermore, in rule-based economies, the institutionalized trust within the legal and regulatory framework reduces the potential for friction between investors with varying experience levels. The transparency and predictability of the legal environment make it easier to align the interests of diverse investors and streamline decision-making processes. A broader range of experiences can also strengthen the strategic oversight of portfolio companies, helping to identify opportunities that might be overlooked by a more homogeneous group of investors. For instance, an investor with significant operational experience may identify strategic risks that could derail the company's path to an exit, while another investor with financial expertise may ensure the business is adequately capitalized for long-term growth. This complementarity increases the likelihood that the company will reach key milestones, positioning it for a successful exit through IPOs or acquisitions.

In contrast, in relationship-based economies like China, a higher variance in the experience of same-round investors may present challenges that reduce the probability of successful exits. While diverse perspectives can provide strategic advantages, in environments where decision-making is often based on personal relationships and informal governance, the increased variance in investor experience can lead to friction. Investors with widely differing experience levels may struggle to align their expectations, leading to conflicts over strategic decisions such as governance structures, market entry strategies or exit timing (Lasserre and Probert, 1994). This discord is particularly problematic in relationship-based economies, where trust and consensus are key to successful collaboration. A highly experienced investor may have well-established relationships and a more hierarchical approach, while a less experienced investor may be more egalitarian and seek a more collaborative decision-making process. These differences can cause delays, misunderstandings and inefficiencies that hinder the progress of the portfolio company toward a successful exit.

Moreover, the lack of strong institutional mechanisms to mediate these differences in experience exacerbates the situation. In the absence of formal contracts or legal enforcement, relational trust is the primary means of resolving conflicts, and a lack of alignment between investors can undermine the entire syndicate. As a result, the increased variance in investor experience could undermine the efficiency and cohesion of the syndicate, making it more difficult for the portfolio company to achieve a successful exit. Therefore, while variance in investor experience can enhance decision-making and governance in rule-based economies, it can introduce significant challenges in relationship-based economies. The lack of institutional mechanisms for conflict resolution and coordination among diverse investors makes it more difficult to manage the complexities of venture growth and exit in such environments:

- H4a.* VCs in strongly rule-based economies (e.g. the USA) will exhibit a higher probability of successful exits as the variance in the experience of same-round investors increases.
- H4b.* VCs in strongly relationship-based economies (e.g. China) will exhibit a lower probability of successful exits as the variance in the experience of same-round investors increases.

4. Sample and method

4.1 Sample

This study investigates how venture capital syndication structures influence exit outcomes in two distinct institutional contexts: the USA and China. The empirical focus is on the semiconductor sector, a capital- and knowledge-intensive industry where investor expertise and coordination are particularly consequential. The sample includes 468 US-based startups and 923 China-based startups, yielding a total of 4,432 and 7,113 investment rounds, respectively. US firms in the data set were founded between 1981 and 2024, while Chinese firms span founding years from 2001 to 2024. These timeframes reflect the emergence and maturation of venture ecosystems in both countries, allowing for meaningful cross-country comparison. The primary data source for this analysis is CB Insights, a proprietary database that provides detailed and verified records of startup financing, investor characteristics and investment outcomes. The platform enables a consistent and granular construction of syndication metrics, capturing both the structure and composition of each investment round.

4.2 Variables

Dependent variables: To test our hypotheses, we measured whether a startup company went public. We created a binary variable to represent this, where 1 indicates an IPO or acquisition by another company, and 0 indicates otherwise (Kim and Park, 2021).

Independent variables: We examine the nature of VC syndication and its impact on exits. To achieve this, we use several VC syndication measures: the total round count of domestic VC syndication, the total round count of global VC syndication, the average investor experience per round and the standard deviation of investor experience per round. The total round count of domestic VC syndication is defined as the number of rounds in which more than one VC participated in the same investment round. The total round count of global VC syndication refers to the number of rounds where more than one VC participated, and at least one of the VCs was from a country different from the one where the target startup's headquarters is located. The average investor experience per round and the standard deviation of investor experience per round capture the characteristics of each investment round. The former represents the average number of prior investments made by each VC in the semiconductor industry before the given investment decision, while the latter reflects the variability in the experience levels of the participating VCs.

Control variables: We control for various factors related to government grants and startup characteristics. Specifically, the nature of government grants is captured through two measures: the total dollar amount of the first government grant and the cumulative total of all government grants received by the startup. The startup's total number of investments reflects the total number of funding rounds secured, regardless of the funding source, while the startup's total investment amount represents the cumulative funding raised (in millions of US\$) across all rounds, including both public and private sources. Startup first investor type identifies the type of the startup's first investor (e.g. government, venture capital, angel investor or corporate). Investor origin country is a categorical variable indicating the country of origin of the investors who funded the startup. Similarly, startup location is a categorical variable that specifies the geographical location of the startup's headquarters at the state level. Startup status is another categorical variable, denoting the current status of the startup (e.g. active, acquired or closed).

4.3 Statistical analysis

To evaluate how VC syndication structures and investor characteristics shape the probability of startup success within different institutional contexts, we adopt a two-stage empirical strategy. Our analysis focuses on comparing the USA and China – two economies with contrasting governance systems and venture financing regimes (Bruton *et al.*, 2005; Wright *et al.*, 2005). First, we estimate a series of logistic regression models to assess the baseline relationships between syndication features and binary exit outcomes (IPO or acquisition). Logistic regression is well-suited for modeling binary dependent variables and allows us to control for a broad set of firm-, investor- and deal-level covariates (Long and Freese, 2014). These include the number of syndicate participants, the presence of foreign (global) investors and the average and heterogeneity of investor experience in each funding round. Control variables account for firm age, cumulative investment, investor type and origin and calendar year fixed effects.

Recognizing the potential endogeneity of key regressors – particularly global syndication and investor experience – we implement instrumental variable Probit (IV-Probit) models as robustness checks. Endogeneity may arise if high-quality startups are more likely to attract global or experienced investors, thereby confounding the direction of causality (Wooldridge, 2010). The IV-Probit framework accommodates binary outcomes while correcting for

endogeneity by leveraging firm-level lagged averages (e.g. of global syndication count) as instruments, following prior empirical approaches in venture capital research (Chemmanur *et al.*, 2011; Hochberg *et al.*, 2007).

We estimate all models separately for US and Chinese subsamples to capture institutional moderation effects. Standard errors are clustered at the firm level to address potential serial correlation. By comparing the IV-Probit estimates with the baseline logit models, we assess the robustness of the observed relationships and the validity of causal claims.

We begin by estimating logistic regression models separately for US and Chinese startups. The dependent variable is a binary indicator equal to 1 if the startup experienced a successful exit (IPO or acquisition), and 0 otherwise. The logistic regression model is specified as:

$$Pr(Y_{it} = 1 | X_{it}) = \frac{1}{1 + \exp(-X_{it}\beta)}$$

where Y_{it} is startup success for firm i at time t , X_{it} is vector of covariates including syndication and investor characteristics, and β is parameter vector to be estimated.

To address potential endogeneity concerns – particularly the possibility that unobserved startup quality may influence both syndication structure and the likelihood of follow-on investment – we use an IV-Probit model using lagged firm-level syndication metrics as instruments. Specifically, we use the average lagged value of global syndication count, computed from past rounds at the firm level, as an instrument for current syndication structure. The assumption underlying this choice is that past syndication reflects syndicate-building norms or firm visibility but does not directly affect the probability of current success except through its influence on syndication structure. While formal over-identification tests are not available in this nonlinear setting, our first-stage results indicate a strong relationship between the instruments and the endogenous regressors, suggesting instrument relevance. The direction and magnitude of second-stage coefficients remain consistent with the logistic models, enhancing confidence in the robustness and causal interpretation of our core findings. We use IV-Probit models, a two-step estimation approach suitable for binary outcomes with endogenous regressors. The IV-Probit model proceeds as follows:

First stage (instrumenting the potentially endogenous regressor Z_{it}):

$$Z_{it} = \delta_0 + \delta_1 \cdot \text{average lagged } Z^{(it)} + \delta_2 \cdot \text{Cit} + u_{it}$$

where average lagged Z is a firm-level average lagged value of the endogenous variable (e.g. global VC syndication and VC syndication)

Second stage (structural Probit model):

$$\Pr(Y_{it} = 1) = \Phi(\beta_0 + \beta_1 \cdot \widehat{Z}_{it} + \gamma C_{it})$$

where $\widehat{Z}_{it}$ is predicted value from the first stage, $\Phi(\cdot)$ is cumulative distribution function of the standard normal distribution.

5. Results

Table 1 presents descriptive statistics and correlations. The analysis includes 4,432 investment observations nested within 468 US companies (Table 1) and 7,113 observations nested within 923 Chinese companies (Table 2). On average, US companies reported a

Table 1. Descriptive statistics and correlations

Variable	Mean	SD	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>US sample: N = 4,432 (468 companies)</i>									
(1) Startup exit (dummy)	0.53	0.50							
(2) Domestic VC syndication count	6.16	3.49	0.087						
(3) Global VC syndication count	0.05	0.35	0.003	-0.150					
(4) Average investor experience by round	8.87	8.53	0.083	-0.166	-0.006				
(5) Standard deviation investor experience by round	8.83	10.71	0.046	-0.012	-0.024	0.847			
(6) Total number of investments	79.27	345.48	-0.024	-0.031	-0.026	-0.026	-0.037		
(7) Total amount of investments	1,026.35	3,897.69	-0.017	-0.029	-0.024	0.004	-0.006	-0.023	
(8) Firm age	23.05	10.24	0.239	0.165	-0.081	-0.232	-0.297	-0.078	-0.206
<i>China sample: N = 7,113 (923 companies)</i>									
(1) Startup exit (dummy)	0.09	0.28							
(2) Domestic VC syndication count	5.66	4.08	0.018						
(3) Global VC syndication count	0.03	0.27	0.059	-0.093					
(4) Average investor experience by round	9.99	9.95	-0.048	0.034	0.027				
(5) Standard deviation investor experience by round	11.29	12.84	-0.041	0.138	0.019	0.909			
(6) Total number of investments	60.82	295.05	0.001	0.001	-0.014	0.057	0.047		
(7) Total amount of investments	1,057.37	3,926.03	0.042	0.039	-0.011	0.067	0.059	-0.010	
(8) Firm age	8.04	5.18	0.409	0.017	0.078	-0.137	-0.099	0.135	-0.016

Note(s): We also controlled for the investor's country, the startup's location city, the startup's current status and year dummies but these are not reported here

Source(s): Authors' own work

Table 2. The results of logistic regressions (US sample – 4,432 / 468 companies, China sample – 7,113 / 923 companies)

Variable	US					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Domestic VC syndication count		0.025 (0.033)				0.045 (0.041)
Global VC syndication count			0.403* (0.178)			0.544* (0.245)
Average investor experience by round				0.004 (0.008)		0.025 (0.029)
Standard deviation investor experience by round					0.016 (0.013)	0.002 (0.022)
Intercept	-1.597* (0.658)	-1.586* (0.663)	-1.675* (0.659)	-1.607* (0.661)	-3.402** (1.307)	-3.672** (1.350)
Total number of investments	-0.012*** (0.003)	-0.012*** (0.003)	-0.012*** (0.003)	-0.013*** (0.004)	-0.001 (0.004)	-0.002 (0.004)
Total amount of investments	0.670*** (0.177)	0.661*** (0.174)	0.671*** (0.177)	0.688*** (0.194)	0.184 (0.188)	0.184 (0.185)
Firm age	0.059*** (0.016)	0.058*** (0.016)	0.060*** (0.016)	0.059*** (0.016)	0.035 (0.025)	0.030 (0.025)
First investor type	0.019 (0.013)	0.018 (0.013)	0.019 (0.013)	0.019 (0.013)	0.023 (0.017)	0.021 (0.017)
Company location ^a	-0.014** (0.005)	-0.014** (0.005)	-0.014** (0.005)	-0.014** (0.005)	-0.014* (0.006)	-0.014* (0.006)
Investor location ^a	0.001 (0.004)	0.001 (0.004)	0.002 (0.004)	0.001 (0.004)	-0.001 (0.005)	0.000 (0.005)
Company status ^a	-0.242* (0.104)	-0.244* (0.103)	-0.239* (0.104)	-0.242* (0.104)	-0.243* (0.101)	-0.245* (0.099)
Investment round ^a	-0.007*** (0.002)	-0.008*** (0.002)	-0.007*** (0.002)	-0.007*** (0.002)	-0.001 (0.003)	-0.002 (0.004)
Log pseudolikelihood	-2,564.4	-2,561.8	-2,554.4	-2,554.7	-2,556.8	-2,536.7
Pseudo R ²	0.163	0.164	0.166	0.166	0.165	0.172
χ^2	131.2***	129.6***	140.9***	131.1***	127.9***	137.2***

Note(s): [†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; – ^aThese variables are identification numbers, and we controlled for them. We also controlled for year dummies but did not report them here
Source(s): Authors' own work

(continued)

Table 2. Continued

Variable	China					
	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Domestic VC syndication count		-0.116** (0.036)				-0.264** (0.085)
Global VC syndication count						-1.109*** (0.386)
Average investor experience by round						0.143* (0.060)
Standard deviation investor experience by round			0.017 (0.232)	-0.019* (0.009)		-0.064† (0.035)
Intercept	-2.997* (1.218)	-3.148* (1.233)	-2.994* (1.220)	-3.184** (1.215)	0.004 (0.016)	-16.91*** (3.910)
Total number of investments	-0.002 (0.003)	-0.002 (0.003)	-0.002 (0.003)	0.001 (0.003)	0.003 (0.005)	-0.003 (0.006)
Total amount of investments	0.065 (0.046)	0.074† (0.042)	0.065 (0.046)	0.067 (0.045)	0.174*** (0.048)	0.181*** (0.037)
Firm age	0.041 (0.032)	0.041 (0.032)	0.041 (0.032)	0.041 (0.032)	0.039 (0.073)	0.059 (0.054)
First investor type	-0.030 (0.021)	-0.021 (0.021)	-0.030 (0.021)	-0.030 (0.021)	0.003 (0.040)	-0.006 (0.043)
Company location ^a	-0.008 (0.006)	-0.010 (0.006)	-0.008 (0.006)	-0.007 (0.006)	-0.032† (0.018)	-0.041* (0.016)
Investor location ^a	0.010 (0.009)	0.010 (0.009)	0.010 (0.009)	0.013 (0.009)	0.022* (0.009)	0.019† (0.010)
Company status ^a	2.364*** (0.191)	2.440*** (0.183)	2.364*** (0.192)	2.376*** (0.191)	3.580*** (0.535)	4.021*** (0.577)
Investment round ^a	-0.032*** (0.003)	-0.030*** (0.003)	-0.032*** (0.003)	-0.032*** (0.003)	-0.016* (0.007)	-0.023* (0.009)
Log pseudolikelihood	-78.34	-73.50	-78.30	-76.99	-78.33	-70.81
Pseudo R ²	0.965	0.967	0.965	0.966	0.965	0.968
χ^2	322.9***	316.9***	345.3***	343.7***	390.6***	383.3***

syndication count of 6.16, reflecting a strong culture of collaboration within a rules-based institutional framework. In contrast, Chinese companies exhibited a slightly lower syndication count of 5.66. Global syndication – where investments involve international collaboration – was relatively rare in both contexts, averaging 0.05 in the USA and 0.03 in China. Differences also emerged in investor experience, with US investors averaging 8.87 years of experience per round, compared to a slightly higher average of 9.99 years in China. Despite these similarities and differences, the success rate of companies transitioning from government funding to private investment was significantly higher in the USA (53%) than in China (9%), underscoring the impact of institutional differences on investment outcomes.

Table 2 presents logistic regression models estimating the likelihood of venture success. We begin with domestic VC syndication count, which represents the number of investors participating in a given round. In the US sample (Models 1–6), this variable is positively signed but not statistically significant ($\beta = 0.025$), offering no support for *H1a*. This result suggests that increasing the number of syndicate partners in the USA neither consistently helps nor hinders startups' prospects for attracting private capital. In contrast, in the Chinese sample (Models 7–12), domestic VC syndication count is negatively and significantly associated with success ($\beta = -0.116$, $p < 0.01$), supporting *H1b*. This finding aligns with institutional theory, which posits that in environments where formal coordination mechanisms are underdeveloped, larger syndicates may create complexity, misaligned interests and diluted accountability – thereby impeding follow-on investment.

Global syndication presents an even starker institutional divergence. In the USA, the number of global VC participants is positively and significantly associated with startup success ($\beta = 0.403$, $p < 0.05$). This supports *H2a* and reflects the value that foreign investors can bring in a mature and legally robust ecosystem, including signaling quality, providing access to international markets and expanding networks. However, in China, global syndication has no significant impacts ($\beta = 0.017$), offering no support for *H2b*. We can see a negative and significant coefficient in Model 12. Although the result is inconsistent depending on the controls. This positive result in the full model may be driven by regulatory restrictions on foreign ownership, weak enforceability of cross-border contracts or limited embeddedness of global investors in China's relational business culture.

The role of investor experience also differs by context. In the USA, average investor experience is positively signed ($\beta = 0.004$) but statistically insignificant, indicating no consistent benefit of investor tenure in a rule-based environment. In China, however, more experienced investors significantly impede startup outcomes ($\beta = -0.019$, $p < 0.05$), showing an opposite to *H3*. But Model 12 shows completely different direction with significance ($\beta = 0.143$, $p < 0.05$). We are not able to support *H3*. Investor heterogeneity, proxied by the standard deviation of investor experience does not affect US outcomes significantly ($\beta = 0.016$) and China outcomes as well ($\beta = 0.004$). Thus, the results show no support for *H4*.

To address concerns of endogeneity – specifically, that more promising startups may attract more investors or global VCs – we reestimate our models using IV-Probit, as shown in Table 3. We instrument syndication variables using firm-level lagged averages to isolate exogenous variation. In the USA, global VC syndication remains a significant and positive predictor of success ($\beta = 0.296$, $p < 0.01$), while domestic VC syndication count continues to be statistically insignificant ($\beta = 0.044$). These results reinforce the conclusion that cross-border VC collaboration is a distinctive strength in the USA, enhancing firm visibility and investor trust without introducing substantial coordination costs, supporting *H2a*. In China, global VC syndication continues to be a negative and statistically significant factor ($\beta = -0.332$, $p < 0.05$), and domestic VC syndication count retains its negative influence

Table 3. The results of IV-Probit models (US sample – 4,432 / 468 companies, China sample – 7,113 / 923 companies)

Variable	US Model 13	China Model 14	US Model 15	China Model 16
Domestic VC syndication count	0.013 (0.025)	-0.024 (0.020)	0.044 (0.042)	-0.063* (0.031)
Global VC syndication count	0.183 (0.225)	0.000 (0.327)	0.296** (0.105)	-0.332* (0.150)
Intercept	-1.188* (0.507)	-5.618*** (0.769)	-1.286* (0.511)	-5.448*** (0.781)
Average investor experience by round	0.023 (0.015)	0.017 (0.012)	0.029+ (0.016)	0.009 (0.010)
Standard deviation investor experience by round	-0.009 (0.013)	-0.014* (0.007)	-0.013 (0.013)	-0.007 (0.008)
Total number of investments	0.000 (0.002)	-0.001 (0.003)	0.001 (0.002)	-0.001 (0.003)
Total amount of investments	0.053 (0.092)	0.043* (0.020)	0.043 (0.093)	0.045* (0.021)
Firm age	0.029*** (0.009)	0.043* (0.019)	0.028** (0.009)	0.044* (0.019)
First investor type	0.016 (0.010)	-0.005 (0.011)	0.015 (0.009)	-0.003 (0.011)
Domestic VC syndication count	-0.010*** (0.003)	-0.004*** (0.001)	-0.597*** (0.154)	-0.885*** (0.182)
Global VC syndication count	0.820*** (0.126)	0.620*** (0.075)	0.829*** (0.034)	0.912*** (0.022)
Average lagged (global) VC syndication count	0.102 (0.067)	0.019 (0.072)	2.002*** (0.513)	0.167 (0.588)
Intercept	0.002 (0.002)	0.000 (0.001)	-0.120*** (0.014)	-0.096*** (0.016)
Average investor experience by round	-0.003+ (0.002)	0.000 (0.001)	0.100*** (0.014)	0.083*** (0.014)
Standard deviation investor experience by round	0.000 (0.000)	-0.000+ (0.000)	-0.005** (0.002)	-0.001 (0.001)
Total number of investments	0.011 (0.009)	0.001 (0.001)	0.192* (0.086)	0.027 (0.017)
Total amount of investments	0.000 (0.001)	0.000 (0.001)	0.005 (0.008)	0.022+ (0.013)
Firm age	-0.001 (0.001)	0.000 (0.000)	-0.009 (0.008)	-0.022** (0.007)
athrho2_1	0.040 (0.074)	-0.079 (0.066)	-0.127 (0.100)	0.233* (0.100)
lnsigma2	-1.278*** (0.122)	-1.541*** (0.099)	0.638*** (0.051)	0.907*** (0.063)
Log pseudolikelihood	-3,306.1	-840.4	-11,689.1	-18,441.9
Wald χ^2	43.8***	490.3***	55.29***	489.6***

Note(s): [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ **Source(s):** Authors' own work

($\beta = -0.063$, $p < 0.05$), confirming the main logistic results and strengthening the causal interpretation of *H1b* and *H2b*.

6. Discussion

This study investigates how institutional contexts shape the structure and outcomes of VC syndication, using a comparative analysis of the semiconductor industries in the USA and China. By focusing on two sharply contrasting institutional frameworks – rules-based in the USA and relationship-based in China – the findings contribute to a nuanced understanding of how institutional conditions affect syndication dynamics, investor coordination and entrepreneurial outcomes.

The empirical results reveal key cross-national differences. In the USA, domestic VC syndication practices were not significantly associated with startup success. This result suggests that in highly institutionalized contexts characterized by strong legal enforcement and regulatory transparency, the marginal value of domestic VC syndication may be diminished. Rather than serving as a compensatory mechanism for institutional voids, syndication in the USA likely plays a complementary role in resource pooling and risk diversification. The presence of efficient formal institutions appears to reduce the dependence on syndicate-driven legitimacy or investor monitoring.

In contrast, domestic VC syndication in China negatively influenced startup success. This finding aligns with the argument that in relationship-based systems, where informal governance and trust substitute for formal enforcement, coordination among multiple investors is more difficult. Misaligned incentives, coordination failures and opaque decision-making processes can erode the potential benefits of syndication. In such settings, the costs of managing syndicate diversity may outweigh the theoretical gains from collaborative investment, particularly in the absence of robust dispute resolution mechanisms.

Global VC syndication yielded divergent results. In the USA, global syndication was positively associated with startup success. US startups benefit from the integration of international investors, leveraging both their financial resources and strategic networks, with institutional protections mitigating cross-border coordination challenges. This reinforces the idea that strong domestic institutions enable firms to internalize the advantages of global capital without incurring excessive transaction costs. In China, however, global VC syndication had no significant effect. The persistence of informal governance mechanisms may limit the effectiveness of foreign partnerships, particularly when cultural and institutional asymmetries hinder trust-building and joint monitoring.

Investor experience also revealed distinct patterns. In China, experienced investors have a mixed impact on startup success. On one hand, they play a critical institutional role by navigating relational governance, resolving conflicts and lending legitimacy to ventures. Their deep embeddedness in informal networks enables them to compensate for weak formal institutions, align interests and facilitate coordination. On the other hand, these investors may operate within closed *guanxi* networks – tight-knit circles based on personal ties – which can limit broader collaboration or access to diverse resources. Such insular relationships may reduce the effectiveness of support for portfolio companies and ultimately hinder their chances of achieving a successful exit. In the USA, however, investor experience did not have a significant effect. This outcome may reflect the leveling effect of institutionalized processes in the US VC ecosystem, where access to information, standard contracts and third-party certifications reduces reliance on investor-specific expertise.

Interestingly, variance in investor experience within a syndicate did not significantly impact outcomes in either country. This suggests that heterogeneous investor backgrounds do not meaningfully enhance – or undermine – startup success once institutional context is

taken into account. In the USA, procedural transparency and standardized governance likely absorb the friction that could result from differences in investor experience. In China, the primacy of trust-based coordination may marginalize the importance of formal credentials, thus muting the effects of variance.

6.1 *Implications for policy*

These findings carry important implications for policymakers seeking to support high-growth entrepreneurship. In the USA, where formal institutions function effectively, policies should continue to encourage global syndication and international collaboration. Strengthening bilateral investment agreements and reducing cross-border regulatory friction could enhance US startups' access to diverse capital sources and global markets.

In China, the results highlight the need to strengthen formal institutions to complement the existing relationship-based systems. Reforms aimed at contract enforcement, transparency and dispute resolution may help reduce coordination failures within syndicates. At the same time, policies that support experienced investors – through tax incentives, coinvestment schemes or government-backed funds – could improve syndicate performance by injecting relational capital and monitoring capacity. Recognizing the importance of informal mechanisms in China's entrepreneurial ecosystem, targeted interventions that bridge formal and informal domains may prove most effective.

These contrasting policy directions reflect the broader distinction between state-led innovation models and market-based ecosystems. The USA supports innovation primarily through competitive market mechanisms, bolstered by programs like small business innovation research/small business technology transfer and institutional frameworks protecting intellectual property. China, by contrast, combines top-down industrial planning with strategic government participation in venture finance. Understanding how these frameworks interact with syndication structures is critical to designing context-appropriate entrepreneurship policies.

6.2 *Implications for practice*

For practitioners, the study underscores the importance of aligning syndication strategies with institutional conditions. In the USA, investors and entrepreneurs can leverage global networks without fearing institutional incompatibility. This allows syndication to function as an innovation amplifier. In China, however, syndicate design must account for the relational nature of governance. Selection of coinvestors based on trust, prior collaboration and informal reputation may be more consequential than their formal credentials. Cross-border syndication in China requires especially careful partner alignment and shared expectations to mitigate the institutional and cultural frictions that global partnerships entail.

6.3 *Limitations and future research directions*

While this study offers novel insights, it also presents several limitations. First, by focusing on the semiconductor industry, the findings may not generalize across sectors. Semiconductors are capital-intensive, state-influenced and long-cycle in nature. Future research should extend this analysis to sectors with different innovation timelines and investor types, such as biotechnology or consumer technology. Second, although CB Insights offers detailed investment data, it may not capture informal or private syndication arrangements, especially in relationship-based environments. Qualitative methods such as interviews with investors, founders or regulators could provide deeper insight into syndicate formation and governance processes. Third, this study captures a cross-sectional view of syndication outcomes. Institutional environments evolve through regulatory reforms,

macroeconomic shocks or geopolitical dynamics. Longitudinal analyses could illuminate how institutional shifts alter the value and structure of syndication over time. Fourth, while the present study provides an outcome-based evaluation of syndication, future research might explore intermediate mechanisms such as learning, mentoring or access to complementary networks. Mixed-method approaches that combine statistical modeling with case studies could unpack the micro-level dynamics behind syndication success or failure. Finally, the role of subnational institutional variation remains underexplored. Both the USA and China exhibit regional differences in legal enforcement, government involvement and venture support ecosystems. Disaggregating the data by region could uncover more nuanced institutional effects and provide location-specific policy guidance.

6.4 Conclusion

This study provides evidence that institutional environments shape not only the structure of venture capital syndication but also its effectiveness. While syndication may offer theoretical benefits – such as resource pooling, diversification and governance – these benefits are deeply contingent on the institutional setting in which they are realized. In the USA, syndication appears to be less critical due to the reliability of formal institutions, whereas in China, syndication introduces coordination costs that can undermine its effectiveness.

By integrating insights from institutional theory with empirical data from two contrasting economies, this research contributes to a more contextualized understanding of venture capital behavior. The findings suggest that institutional environments must be explicitly considered in both theory development and policy design related to entrepreneurial finance. For practitioners and policymakers alike, aligning syndication strategies with institutional realities is essential to fostering effective investment and innovation outcomes.

References

- Aeeni, Z., Motavaseli, M., Sakhdari, K. and Saeedikiya, M. (2019), “Extending the potential of Baumol’s entrepreneurial allocation theory: toward the entrepreneur-institution nexus”, *Journal of Entrepreneurship in Emerging Economies*, Vol. 11 No. 3, pp. 416-435.
- Aeeni, Z., Saeedikiya, M., Sakhdari, K. and Sadeghi, V.J. (2024), “Blooming in the cracks: productive entrepreneurship amid institutional voids”, *Small Business Economics*, Vol. 64 No. 4, pp. 1723-1762.
- Ahlstrom, D., Bruton, G.D. and Yeh, K.S. (2008), “Private firms in China: building legitimacy in an emerging economy”, *Journal of World Business*, Vol. 43 No. 4, pp. 385-399.
- Baumol, W.J. (1996), “Entrepreneurship: productive, unproductive, and destructive”, *Journal of Business Venturing*, Vol. 11 No. 1, pp. 3-22.
- Baumol, W.J., Litan, R.E., and Schramm, C.J. (2007), *Good Capitalism, Bad Capitalism, and the Economics of Growth and Prosperity*, Yale University Press.
- Brander, J.A., Amit, R. and Antweiler, W. (2002), “Venture-capital syndication: improved venture selection vs. the value-added hypothesis”, *Journal of Economics and Management Strategy*, Vol. 11 No. 3, pp. 423-452.
- Bruton, G.D. and Ahlstrom, D. (2003), “An institutional view of China’s venture capital industry: explaining the differences between China and the West”, *Journal of Business Venturing*, Vol. 18 No. 2, pp. 233-259.
- Bruton, G.D., Ahlstrom, D. and Li, H. (2010), “Institutional theory and entrepreneurship: where are we now and where do we need to move in the future?”, *Entrepreneurship Theory and Practice*, Vol. 34 No. 3, pp. 421-440.

- Bruton, G.D., Ahlstrom, D. and Puky, T. (2009), "Institutional differences and the development of entrepreneurial ventures: a comparison of the venture capital industries in Latin America and Asia", *Journal of International Business Studies*, Vol. 40 No. 5, pp. 762-778.
- Bruton, G.D., Fried, V.H. and Manigart, S. (2005), "Institutional influences on the worldwide expansion of venture capital", *Entrepreneurship Theory and Practice*, Vol. 29 No. 6, pp. 737-760.
- Bygrave, W., Fast, N., Khoylian, R., Vincent, L., and Yue, W. (2022), "Early rates of return of 131 venture capital funds started 1978–1984", *Venture Capital*, Routledge, pp. 449-461.
- Chemmanur, T.J., Krishnan, K. and Nandy, D.K. (2011), "How does venture capital financing improve efficiency in private firms? A look beneath the surface", *Journal of Financial Economics*, Vol. 99 No. 1, pp. 45-73.
- Cumming, D.J. and MacIntosh, J.G. (2006), "Crowding out private equity: Canadian evidence", *Journal of Business Venturing*, Vol. 21 No. 5, pp. 569-609.
- Cumming, D. and Dai, N. (2010), "Local bias in venture capital investments", *Journal of Empirical Finance*, Vol. 17 No. 3, pp. 362-380.
- Dai, N., Jo, H. and Kassicieh, S. (2012), "Cross-border venture capital investments in Asia: selection and exit performance", *Journal of Business Venturing*, Vol. 27 No. 6, pp. 666-684.
- DiMaggio, P.J. (1988), "Interest and agency", in "Institutional theory", *Institutional Patterns and Organizations: Culture and Environment*, pp. 3-22.
- DiMaggio, P.J. and Powell, W.W. (1983), "The iron cage revisited: institutional isomorphism and collective rationality in organizational fields", *American Sociological Review*, Vol. 48 No. 2, pp. 147-160.
- Dimov, D. and Milanov, H. (2010), "The interplay of need and opportunity in venture capital investment syndication", *Journal of Business Venturing*, Vol. 25 No. 4, pp. 331-348.
- Falconieri, S., Filatotchev, I. and Tastan, M. (2019), "Size and diversity in VC syndicates and their impact on IPO performance", *The European Journal of Finance*, Vol. 25 No. 11, pp. 1032-1053.
- Folta, P.H. (1999), "The rise of venture capital in China", *China Business Review*, Vol. 26 No. 6, pp. 26-30.
- Foster, R.N. (1986), "Working the S-curve: assessing technological threats", *Research Management*, Vol. 29 No. 4, pp. 17-20.
- Fukuyama, F. (1996), *Trust: The Social Virtues and the Creation of Prosperity*, Simon and Schuster.
- Garud, R., Jain, S. and Kumaraswamy, A. (2002), "Institutional entrepreneurship in the sponsorship of common technological standards: the case of Sun Microsystems and Java", *Academy of Management Journal*, Vol. 45 No. 1, pp. 196-214.
- George, B., Erikson, T. and Parhankangas, A. (2016), "Preventing dysfunctional conflict: examining the relationship between different types of managerial conflict in venture capital-backed firms", *Venture Capital*, Vol. 18 No. 4, pp. 279-296.
- Gnyawali, D.R. and Fogel, D.S. (1994), "Environments for entrepreneurship development: key dimensions and research implications", *Entrepreneurship Theory and Practice*, Vol. 18 No. 4, pp. 43-62.
- Gompers, P.A. and Lerner, J. (2001), "The venture capital revolution", *Journal of Economic Perspectives*, Vol. 15 No. 2, pp. 145-168.
- Gompers, P.A., and Lerner, J. (2004), *The Venture Capital Cycle*, 2nd ed., MIT Press.
- Hall, J. and Hofer, C.W. (1993), "Venture capitalists' decision criteria in new venture evaluation", *Journal of Business Venturing*, Vol. 8 No. 1, pp. 25-42.
- Harrison, R.T. and Mason, C.M. (2000), "Venture capital market complementarities: the links between business angels and venture capital funds in the United Kingdom", *Venture Capital*, Vol. 2 No. 3, pp. 223-242.
- Heidl, R.A., Steensma, H.K. and Phelps, C. (2014), "Divisive faultlines and the unplanned dissolutions of multipartner alliances", *Organization Science*, Vol. 25 No. 5, pp. 1351-1371.

- Hellmann, T. and Puri, M. (2002), "Venture capital and the professionalization of start-up firms: empirical evidence", *The Journal of Finance*, Vol. 57 No. 1, pp. 169-197.
- Hochberg, Y.V., Lindsey, L.A. and Westerfield, M.M. (2015), "Resource accumulation through economic ties: evidence from venture capital", *Journal of Financial Economics*, Vol. 118 No. 2, pp. 245-267.
- Hochberg, Y.V., Ljungqvist, A. and Lu, Y. (2007), "Whom you know matters: venture capital networks and investment performance", *The Journal of Finance*, Vol. 62 No. 1, pp. 251-301.
- Jensen, M.C. and Meckling, W.H. (1976), "Theory of the firm: managerial behavior, agency costs and ownership structure", *Journal of Financial Economics*, Vol. 3 No. 4, pp. 305-360.
- Khurshed, A., Mohamed, A., Schwienbacher, A. and Wang, F. (2020), "Do venture capital firms benefit from international syndicates?", *Journal of International Business Studies*, Vol. 51 No. 6, pp. 986-1007.
- Kim, J.Y. and Park, H.D. (2021), "The influence of venture capital syndicate size on venture performance", *Venture Capital*, Vol. 23 No. 2, pp. 179-203.
- Lasserre, P. and Probert, J. (1994), "Competing on the Pacific Rim: high risks and high returns", *Long Range Planning*, Vol. 27 No. 2, pp. 12-35.
- Lawrence, T.B., Hardy, C. and Phillips, N. (2002), "Institutional effects of interorganizational collaboration: the emergence of proto-institutions", *Academy of Management Journal*, Vol. 45 No. 1, pp. 281-290.
- Lerner, J. (1994), "Venture capitalists and the decision to go public", *Journal of Financial Economics*, Vol. 35 No. 3, pp. 293-316.
- Long, J.S. and Freese, J. (2014), *Regression Models for Categorical Dependent Variables Using Stata*, 3rd ed., Stata Press, College Station, TX.
- Manigart, S., Lockett, A., Meuleman, M., Wright, M., Landström, H. and *et al.* (2006), "Venture capitalists' decision to syndicate", *Entrepreneurship Theory and Practice*, Vol. 30 No. 2, pp. 131-153.
- Meggison, W.L. and Weiss, K.A. (1991), "Venture capital certification in initial public offerings", *The Journal of Finance*, Vol. 46 No. 3, pp. 879-903.
- Meyer, J.W. and Rowan, B. (1977), "Institutionalized organizations: formal structure as myth and ceremony", *American Journal of Sociology*, Vol. 83 No. 2, pp. 340-363.
- North, D.C. (1990), *Institutions, Institutional Change and Economic Performance*, Cambridge University Press.
- Peng, M.W. (2003), "Institutional transitions and strategic choices", *The Academy of Management Review*, Vol. 28 No. 2, pp. 275-296.
- Peng, M.W., Wang, D.Y.L. and Jiang, Y. (2008), "An institution-based view of international business strategy: a focus on emerging economies", *Journal of International Business Studies*, Vol. 39 No. 5, pp. 920-936.
- Poppo, L., Zhou, K.Z. and Rhu, S. (2008), "Alternative origins to interorganizational trust: an interdependence perspective on the shadow of the past and the shadow of the future", *Organization Science*, Vol. 19 No. 1, pp. 39-55.
- Rajan, R.G. and Zingales, L. (2003), "The great reversals: the politics of financial development in the twentieth century", *Journal of Financial Economics*, Vol. 69 No. 1, pp. 5-50.
- Rao, H., Morrill, C. and Zald, M.N. (2000), "Power plays: how social movements and collective action create new organizational forms", *Research in Organizational Behavior*, Vol. 22, pp. 237-281.
- Sapienza, H.J. and Korsgaard, M.A. (1996), "Procedural justice in entrepreneur-investor relations", *Academy of Management Journal*, Vol. 39 No. 3, pp. 544-574.
- Scott, W.R. (2013), *Institutions and Organizations: Ideas, Interests, and Identities*, 4th ed., Sage Publications.

- Sorenson, O. and Stuart, T.E. (2001), "Syndication networks and the spatial distribution of venture capital investments", *American Journal of Sociology*, Vol. 106 No. 6, pp. 1546-1588.
- Soto, H. (2013), *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*, Basic Books.
- Stillman, P.G. (1974), "The concept of legitimacy", *Polity*, Vol. 7 No. 1, pp. 32-56.
- Tian, X. (2011), "The causes and consequences of venture capital stage financing", *Journal of Financial Economics*, Vol. 101 No. 1, pp. 132-159.
- Wang, L. and Wang, S. (2011), "Cross-border venture capital performance: evidence from China", *Pacific-Basin Finance Journal*, Vol. 19 No. 1, pp. 71-97.
- White, S., Gao, J. and Zhang, W. (2002), "Financing new ventures in China: system antecedents and institutionalization", *Research Policy*, Vol. 34 No. 1, pp. 75-93.
- Williamson, O.E. (1985), *The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting*, Free Press.
- Wooldridge, J.M. (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, Cambridge, MA.
- Wright, M. and Lockett, A. (2003), "The structure and management of alliances: syndication in the venture capital industry", *Journal of Management Studies*, Vol. 40 No. 8, pp. 2073-2102.
- Wright, M., Pruthi, S. and Lockett, A. (2005), "International venture capital research: from cross-country comparisons to crossing borders", *Entrepreneurship Theory and Practice*, Vol. 29, No. 6, pp. 681-698.
- Zacharakis, A.L. and Meyer, G.D. (1998), "A lack of insight: do venture capitalists really understand their own decision process?", *Journal of Business Venturing*, Vol. 13 No. 1, pp. 57-76.
- Zhang, L., Gupta, A.K. and Hallen, B.L. (2017), "The conditional importance of prior ties: a group-level analysis of venture capital syndication", *Academy of Management Journal*, Vol. 60 No. 4, pp. 1360-1386.

Further reading

- He, C., Lu, J. and Qian, H. (2019), "Entrepreneurship in China", *Small Business Economics*, Vol. 52 No. 3, pp. 563-572.
- Kang, S. (2019), "The impact of corporate venture capital involvement in syndicates", *Management Decision*, Vol. 57 No. 1, pp. 131-151.
- Mitchell, W. (1989), "Whether and when? Probability and timing of incumbents' entry into emerging industrial subfields", *Administrative Science Quarterly*, Vol. 34 No. 2, pp. 208-230.

About the authors

KwangWook Gang holds a PhD in strategic management from Rensselaer Polytechnic Institute, Troy, NY, USA. Currently serving as an Associate Professor at the Franklin P. Perdue School of Business, Salisbury University, MD, USA, he specializes in researching a firm's entrepreneurial activities and technological strategies within the evolving landscape of industries, with a particular focus on disruptive technology. His academic pursuits also extend to exploring a firm's product strategy, growth dynamics and corporate governance in emerging economies. Noteworthy is his contribution to esteemed publications such as *Technological Forecasting and Social Change*, *Technology Analysis and Strategic Management*, *IEEE Transactions on Engineering Management* and various other peer-reviewed journals. KwangWook Gang is the corresponding author and can be contacted at: kxgang@salisbury.edu

Yonggi Kim graduated from Seoul National University, Korea with a PhD in Environmental Planning with a focus on economic and strategic behavior of firms. Currently serving as an Associate Research Fellow at the Science and Technology Policy Institute, Sejong, Korea, he focuses on the interplay between national technology policy and the changing industrial environment, with an

emphasis on critical emerging technologies. His academic pursuits also extend to the study of the industry dynamics of deep-tech startups, the global ecosystem of the deep-tech industry and the competition for technological hegemony. Notably, he has contributed to national policy studies such as research on improving the burden criteria for government-funded R&D expenditures, a study on the strategy for attracting advanced technology companies, and various other policy studies.

Byunghoon Kim (Member, IEEE) received his PhD degree in Industrial and Systems Engineering from Rutgers University, New Brunswick, NJ, USA, in 2015. He is currently an Associate Professor with the Department of Industrial and Management Engineering, Hanyang University, Ansan, South Korea. Starting in Fall 2024, he is a visiting scholar at Rutgers University. His research interests include data mining in semiconductor manufacturing processes, network data analysis and high-dimensional data analysis.

Seungbeom Kim is an Associate Professor in the School of Business Administration at Hongik University, Seoul, South Korea. He earned his PhD in Business Administration from the University of Southern California, Los Angeles, CA, USA. His research interests include service operations, revenue management with a focus on customer psychology, supply chain management and technology management.